

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	CSA0615	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO4: Articulation (BL4, BL5)	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	20% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
Student Name:	V.LASYA	Reg. No.:	192524186
Section / Batch:	B.TECH AIDS	Date:	28-08-2026

Instructions to Students

- Answer both questions. Each question carries 50 Marks. Total: 100 Marks.
- Carefully analyze the given questions.
- Explain in detail with sample code if needed.
- Clearly identifies all key issues.
- Thorough analysis with validated results and performance evaluation

Question Type	Description	Questions	Total Marks
Industry Problem Solving Task	Focus on Solving optimization problems using dynamic programming and greedy strategies	Q1 – Q2	100

SECTION A – Industry Problem Solving Task

Code of Conduct

I ____ V.LASYA(192524186)____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the Student: _____

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QUESTIONS

Q1

Greedy Technique
30 marks

Q2

Greedy Technique
50 marks

Q2 Greedy Technique

50 marks

Smart Healthcare Resource Allocation (Optimization + Scheduling)

A multi-specialty hospital must allocate limited ICU beds and medical staff among critical patients. Analyze how optimization algorithms can improve resource utilization. Identify constraints such as patient priority, emergency arrivals, and staff availability. Design a scheduling model using efficient algorithms. Discuss feasibility in real-time healthcare systems. Evaluate performance based on response time and utilization. Interpret results in improving patient care efficiency.

Your Code

```
1 n=int(input())
2 beds=int(input())
3 patients=[]
4 for i in range(n):
5     p=int(input())
6     patients.append((p,i+1))
7 patients.sort(reverse=True)
8 for i in range(n):
9     if i<beds:
10        print("patient",patients[i][1],"allocated")
11    else:
12        print("patient",patients[i][1],"waiting")
13
```

Input

5
2
5
3

Output

```
patient 3 allocated
patient 5 allocated
patient 1 waiting
patient 2 waiting
patient 4 waiting
```

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QUESTIONS

Q1

Greedy Technique
50 marks

Q2

Greedy Technique
50 marks

Q2 Greedy Technique

50 marks

Financial Portfolio Optimization (Dynamic Programming + Greedy)

An investment firm must allocate funds across assets to maximize returns while minimizing risk. Analyze how optimization algorithms can model this problem. Identify constraints such as budget limits and risk tolerance. Design a solution combining dynamic programming and greedy methods. Discuss feasibility in real-time trading systems. Evaluate efficiency and performance. Interpret results in financial decision-making.

Your Code

```

1 n=int(input())
2 budget=int(input())
3 assets=[]
4 for i in range(n):
5     cost,profit=map(int,input().split())
6     assets.append((profit/cost,cost,profit,i+1))
7 assets.sort(reverse= True)
8 total=0
9 for ratio,cost,profit,asset in assets:
10     if cost<=budget:
11         budget-=cost
12         total+=profit
13         print("asset",asset,"selected")
14 print("maximum return:",total)

```

Run

Save

Submit

Input

```

20 40
30 60
50 75
40 80

```

Output

```

asset 4 selected
asset 2 selected
asset 1 selected
maximum return: 180

```

RUBRICS FOR CALCULATION

Industry Problem Solving Task

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Mark Evaluation:

Case Study	Understanding of Industry Problem (20)	Application of Engineering Concepts (25)	Solution Design (25)	Use of Tools (15)	Analysis (10)	Professionalism (5)	Total Marks (Each - 50)
Q1							
Q2							
	Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Name: _____ Signature: _____	Signature: _____	Signature: _____

Date: _____	Date: _____	Date: _____
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